

A finite-dimensional signed L^1 uncertainty inequality on the Boolean cube

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Abstract

Polyanskiy and Samorodnitsky conjectured that their sharp asymptotic L^1 uncertainty estimate for nonnegative low-degree functions on the Boolean cube remains true for arbitrary real functions. We prove the stronger finite-dimensional inequality

$$\|f\|_\infty \leq \sqrt{\frac{\binom{2n}{d}}{\binom{2n}{n}}} \sum_{x \in \{-1,1\}^n} |f(x)|, \quad \deg f \leq d.$$

Its logarithm is $-n[\log 2 - h(d/(2n))] + O(\log n)$, exactly the conjectured exponent. The proof translates a maximizer to the all-ones vertex, symmetrizes over coordinates, and uses the exact Christoffel function $K_d(0,0) = \binom{2n}{d}$ for the squared binomial (Hahn) measure.

1 Source conjecture

Let $h(t) = -t \log t - (1-t) \log(1-t)$ be binary entropy in natural units. After proving an L^2 uncertainty principle, Polyanskiy and Samorodnitsky compare it with a result for nonnegative functions and state the following conjecture on page 20 of the published manuscript [1]:

Theorem 11 shows that any low-degree polynomial (restricted to the hypercube) smears its L_2 -norm so evenly that one needs to sum $e^{nR_{LP1}(\delta)}$ top values in order to obtain a sizable fraction of its overall L_2 -norm. It is interesting to compare this with [34] showing that any f that is a) a degree $\leq \delta n$ polynomial and b) $f \geq 0$ satisfies

$$\max_{x \in \{\pm 1\}^n} |f(x)| \leq e^{-n(\ln 2 - h(\frac{\delta}{2})) + o(n)} \sum_{x \in \{\pm 1\}^n} |f(x)|. \quad (70)$$

We *conjecture* that (70) holds for all $f : \{\pm 1\}^n \rightarrow \mathbb{R}$ of degree $\leq \delta n$. This could be called an L_1 -version of the uncertainty principle. If true, it would imply that the sum of any $e^{nR_{Ham}(\delta)}$, $R_{Ham}(\delta) = \ln 2 - h(\frac{\delta}{2})$ values of $|f(x)|$ is negligible compared to the sum over all of $\{\pm 1\}^n$.

In exact mathematical form, their equation (70) says that, for $f \geq 0$ of Fourier–Walsh degree at most δn ,

$$\max_{x \in \{-1,1\}^n} |f(x)| \leq \exp\left(-n \left[\log 2 - h\left(\frac{\delta}{2}\right)\right] + o(n)\right) \sum_{x \in \{-1,1\}^n} |f(x)|. \quad (1)$$

The immediately following sentence is:

“We conjecture that (70) holds for all $f : \{\pm 1\}^n \rightarrow \mathbb{R}$ of degree $\leq \delta n$.”

The result below proves this statement and replaces the unspecified $o(n)$ by an explicit $O(\log n)$ remainder.

2 Definitions and proof intuition

Write $\chi_S(x) = \prod_{i \in S} x_i$. The Fourier–Walsh degree of $f : \{-1, 1\}^n \rightarrow \mathbb{R}$ is the largest $|S|$ for which the coefficient of χ_S is nonzero. All L^1 norms in the theorem use counting measure:

$$\|f\|_{\ell^1} = \sum_{x \in \{-1, 1\}^n} |f(x)|, \quad \|f\|_{\infty} = \max_{x \in \{-1, 1\}^n} |f(x)|.$$

The proof has two independent ideas. First, translate a point attaining $\|f\|_{\infty}$ to $(1, \dots, 1)$ and average over all coordinate permutations. Translation preserves Fourier degree, the average keeps the chosen value, and convexity makes its ℓ^1 norm no larger. The averaged function depends only on the number k of negative coordinates, hence is a univariate polynomial $P(k)$ of degree at most d .

Second, the k -distribution of a uniform cube point is binomial. Squaring this binomial weight produces a Hahn orthogonality measure. Its degree- d Christoffel function at the endpoint is exactly $\binom{2n}{d}$; one Cauchy–Schwarz inequality then bounds $P(0)$ by the binomially weighted L^1 norm of P .

3 The exact theorem

Theorem 1 (signed L^1 uncertainty). *Let $0 \leq d \leq n$. Every real function $f : \{-1, 1\}^n \rightarrow \mathbb{R}$ of Fourier–Walsh degree at most d satisfies*

$$\|f\|_{\infty} \leq \sqrt{\frac{\binom{2n}{d}}{\binom{2n}{n}}} \|f\|_{\ell^1}. \quad (2)$$

Corollary 2 (the source conjecture). *Uniformly for $0 \leq d \leq n$,*

$$\sqrt{\frac{\binom{2n}{d}}{\binom{2n}{n}}} = \exp\left(-n \left[\log 2 - h\left(\frac{d}{2n}\right)\right] + O(\log n)\right). \quad (3)$$

Consequently, if $d \leq \delta n$, then (1) holds for arbitrary real f .

Proof. The uniform estimate $\log \binom{N}{k} = Nh(k/N) + O(\log N)$ gives (3). For $d \leq \delta n \leq n$, monotonicity of $\binom{2n}{d}$ in this range completes the deduction from Theorem 1. $\square$

4 The squared-binomial Christoffel identity

For $0 \leq k \leq n$, put

$$\mu_k = 2^{-n} \binom{n}{k}, \quad Z = \sum_{k=0}^n \mu_k^2 = 4^{-n} \binom{2n}{n}, \quad \nu_k = \frac{\mu_k^2}{Z} = \frac{\binom{n}{k}^2}{\binom{2n}{n}}.$$

The identity for Z is Vandermonde’s convolution.

For a probability measure ν on $\{0, \dots, n\}$, let

$$K_d^\nu(0, 0) = \sup_{0 \neq P, \deg P \leq d} \frac{|P(0)|^2}{\sum_{k=0}^n \nu_k |P(k)|^2}$$

be its degree- d Christoffel function at zero.

Lemma 3 (endpoint Hahn identity). *For the squared-binomial probability measure above,*

$$K_d^\nu(0, 0) = \binom{2n}{d}, \quad 0 \leq d \leq n.$$

Proof. We record the short normalization calculation so that no asymptotic orthogonal-polynomial result is hidden in the proof. With $(a)_r$ denoting the rising factorial, define the terminating Hahn polynomials

$$H_j(k) = {}_3F_2 \left(\begin{matrix} -j, j-2n-1, -k \\ -n, -n \end{matrix}; 1 \right) = \sum_{r=0}^j \frac{(-j)_r (j-2n-1)_r (-k)_r}{(-n)_{r!}^2}.$$

They have degree j and $H_j(0) = 1$. Expanding the terminating sums and applying Vandermonde's convolution (equivalently, the standard finite Hahn orthogonality calculation [2, Sec. 9.5]) gives

$$\sum_{k=0}^n \nu_k H_i(k) H_j(k) = \frac{\mathbf{1}_{\{i=j\}}}{\binom{2n}{j} - \binom{2n}{j-1}}, \quad (4)$$

where $\binom{2n}{-1} = 0$.

Thus $H_0, \dots, H_d$ are an orthogonal basis of the polynomials of degree at most d . The reproducing-kernel formula and $H_j(0) = 1$ yield

$$K_d^\nu(0, 0) = \sum_{j=0}^d \frac{|H_j(0)|^2}{\|H_j\|_{L^2(\nu)}^2} = \sum_{j=0}^d \left[\binom{2n}{j} - \binom{2n}{j-1} \right] = \binom{2n}{d}.$$

This also proves the claimed normalization directly. The accompanying exact-arithmetic script expands (4) for all degrees through $n = 12$ as an independent check. $\square$

Lemma 4 (radial L^1 evaluation). *Every real polynomial P of degree at most $d \leq n$ satisfies*

$$|P(0)| \leq \sqrt{\frac{\binom{2n}{d}}{Z}} \sum_{k=0}^n \mu_k |P(k)|. \quad (5)$$

Proof. Set $A = \sum_k \mu_k |P(k)|$. Since all summands are nonnegative,

$$A^2 \geq \sum_{k=0}^n \mu_k^2 |P(k)|^2 = Z \|P\|_{L^2(\nu)}^2.$$

The definition of the Christoffel function and Lemma 3 give

$$|P(0)|^2 \leq \binom{2n}{d} \|P\|_{L^2(\nu)}^2 \leq \frac{\binom{2n}{d}}{Z} A^2.$$

Taking square roots proves (5). $\square$

5 Proof of the theorem

Proof of Theorem 1. Choose $x^0 \in \{-1, 1\}^n$ with $|f(x^0)| = \|f\|_\infty$ and multiply f by a sign. The cube translation

$$g(x) = f(x_1^0 x_1, \dots, x_n^0 x_n)$$

has the same Fourier degree and the same ℓ^1 norm, and $g(1, \dots, 1) = \|f\|_\infty$.

Average g over the symmetric group:

$$\bar{g}(x) = \frac{1}{n!} \sum_{\pi \in S_n} g(x_{\pi(1)}, \dots, x_{\pi(n)}).$$

Then $\deg \bar{g} \leq d$, the all-ones vertex is fixed, and Jensen's inequality followed by permutation invariance of counting measure gives

$$\bar{g}(1, \dots, 1) = \|f\|_\infty, \quad \|\bar{g}\|_{\ell^1} \leq \|g\|_{\ell^1} = \|f\|_{\ell^1}. \quad (6)$$

A symmetric Fourier–Walsh polynomial of degree at most d is a linear combination of the first d Krawtchouk polynomials. Hence $\bar{g}(x) = P(k)$, where $k = |\{i : x_i = -1\}|$ and P is a univariate polynomial of degree at most d . In particular,

$$\sum_{k=0}^n \mu_k |P(k)| = 2^{-n} \|\bar{g}\|_{\ell^1}, \quad P(0) = \|f\|_\infty.$$

Lemma 4, the value $Z = 4^{-n} \binom{2n}{n}$, and (6) now give

$$\begin{aligned} \|f\|_\infty &\leq \sqrt{\frac{\binom{2n}{d}}{4^{-n} \binom{2n}{n}}} 2^{-n} \|\bar{g}\|_{\ell^1} \\ &\leq \sqrt{\frac{\binom{2n}{d}}{\binom{2n}{n}}} \|f\|_{\ell^1}, \end{aligned}$$

which is (2). □

6 Verification, novelty, and limitations

There are several quick checks on the exact constant. If $d = n$, it is one and (2) is the trivial inequality $\|f\|_\infty \leq \|f\|_1$. If $d = 0$, it differs from the sharp constant 2^{-n} only by the polynomial factor $4^{n/2} / \sqrt{\binom{2n}{n}} = O(n^{1/4})$, so it has the correct exponential scale. Exact symbolic computation verifies the Hahn orthogonality, reciprocal norms, and telescoping identity for every $1 \leq n \leq 12$; see `verify_hahn_identity.py`.

Targeted searches of the source, its published version, the run registry, Boolean-cube Nikolskii inequalities, the exact binomial-ratio constant, and squared-binomial Christoffel functions found no prior statement of Theorem 1. The source TeX contains an appendix-only working note mentioning a signed permutation-invariant estimate, and Samorodnitsky's older preprint [3] develops the relevant squared-binomial/Hahn machinery. Those are consistent with Lemma 4; neither located source states the translation–symmetrization reduction or the finite inequality (2). Literature novelty is nevertheless the main point requiring expert review.

The theorem resolves only the signed L^1 conjecture following equation (70). It does not solve the source's separate conjecture that Hamming balls are asymptotically worst for support-constrained hypercontractivity, nor its more structured uncertainty questions.

Human review recommendation. Before dissemination, verify the normalization in (4), the degree preservation in the radial reduction, and novelty against the literature on Hahn Christoffel functions and discrete Nikolskii inequalities.

References

- [1] Y. Polyanskiy and A. Samorodnitsky, *Improved log-Sobolev inequalities, hypercontractivity and uncertainty principle on the hypercube*, J. Funct. Anal. 277 (2019), 108280; arXiv:1606.07491.
- [2] R. Koekoek, P. A. Lesky, and R. F. Swarttouw, *Hypergeometric Orthogonal Polynomials and Their q -Analogues*, Springer, 2010.
- [3] A. Samorodnitsky, *Extremal properties of solutions for Delsarte's linear program*, preliminary manuscript, 1998.